

СУБД

Лекция 7

TRANSLATE

- `SELECT TRANSLATE ('abc1cba', 'cba', 'def')`
`FROM DUAL;`
- `fe1d1def`

DECODE

- `SELECT DECODE ('equal', 'equal', 'TRUE','FALSE') FROM DUAL`
- TRUE
- `SELECT DECODE ('equal', 'unequal', 'TRUE', 'FALSE') FROM DUAL`
- FALSE

DECODE

```
SELECT DECODE (column1,  
1,'lecture',  
2,'practice',  
3,'labs',  
'Self education')  
FROM subjects
```

ROLLUP

- SELECT department_id, job_id, sum(salary)
- FROM employees
- GROUP BY ROLLUP(department_id, job_id)

CUBE

- SELECT department_id, job_id, sum(salary)
- FROM employees
- GROUP BY CUBE(department_id, job_id)

GROUPING

- SELECT GROUPING(department_id),
department_id, job_id, sum(salary)
- FROM employees
- GROUP BY CUBE(department_id,job_id)

-
- SELECT
DECODE(GROUPING(department_id),1,'Итого',
department_id) as dep_id, sum(salary)
 - FROM employees
 - GROUP BY ROLLUP(department_id)

GROUPING SETS

```
SELECT department_id, job_id, SUM (salary)
FROM hr.employees
GROUP BY GROUPING SETS (department_id,
job_id)
```

Rownum

```
SELECT rownum, last_name  
FROM employees  
ORDER BY last_name
```

Оконные функции

- OVER()

RANK(), DENSE_RANK

```
SELECT product_type, sum(amount),  
RANK() OVER (ORDER BY SUM(amount) DESC),  
DENSE_RANK() OVER (ORDER BY  
SUM(amount) DESC)  
FROM sales  
GROUP BY product_type  
ORDER BY product_type
```

NULLS LAST и NULLS FIRST

```
SELECT product_type, sum(amount),  
RANK() OVER (ORDER BY SUM(amount) DESC  
NULLS LAST),  
DENSE_RANK() OVER (ORDER BY SUM(amount)  
DESC NULLS FIRST)  
FROM sales  
GROUP BY product_type  
ORDER BY product_type
```

LAG и LEAD

```
SELECT res_year, avg(amount),  
LAG( avg(amount),1) OVER (order by res_year),  
LEAD( avg(amount),1) OVER (order by res_year)  
FROM sales  
  
GROUP BY res_year  
ORDER BY res_year
```

PARTITION BY

```
SELECT ROW_NUMBER () OVER (PARTITION  
BY department_id ORDER BY last_name),  
       last_name,  
       department_id  
FROM hr.employees e1  
ORDER BY department_id DESC NULLS LAST
```

Скользящее окно

- {ROWS / RANGE}
BETWEEN
{ {UNBOUNDED PRECEDING / CURRENT ROW /
{UNBOUNDED / выражение 1} {PRECEDING /
FOLLOWING}}
AND
{ {UNBOUNDED FOLLOWING / CURRENT ROW /
{UNBOUNDED / выражение 2} {PRECEDING /
FOLLOWING}}

- `SELECT` `ename`, `deptno`, `job`,
`SUM(sal)`
`OVER (PARTITION BY deptno, job ORDER BY`
`hiredate`
`ROWS BETWEEN UNBOUNDED`
`PRECEDING AND CURRENT ROW)` `sum_sal`,
- **`Avg(sal)`** `OVER (ORDER BY hiredate`
`ROWS BETWEEN 2 PRECEDING AND`
`CURRENT ROW)` `as avg_sal`
`FROM emp`;

FIRST_VALUE и LAST_VALUE

- SELECT ename, hiredate, sal,
FIRST_VALUE(sal)
OVER (ORDER BY hiredate
ROWS BETWEEN 2 PRECEDING AND
CURRENT ROW) first_rows
- From emp

RATIO_TO_REPORT

```
SELECT last_name,  
       SUM (salary),  
       RATIO_TO_REPORT (SUM (salary)) OVER  
(PARTITION BY department_id)  
FROM hr.employees e1  
WHERE e1.department_id = 90  
GROUP BY last_name, department_id
```

MODEL

- PARTITION BY – правила секционирования
- DIMENSION BY – указывает на измерения в массиве
- MEASURES – указывает на ячейки и название массива

MODEL

```
SELECT product_type,year, month,sales_amount
FROM sales
MODEL
PARTITION BY product_type
DIMENSION BY (month,year)
MEASURES(sales_amount) (
Sales_amount[1,2021]=sales_amount[1,2020]*1.3,
Sales_amount[2,2021]=sales_amount[2,2020]+sales_amount[1,2020],
Sales_amount[3,2021]=ROUND(AVG(sales_amount)[month BETWEEN 1
AND 3],2))
```

MODEL

Sales_amount[FOR month FROM 1 TO 3
INCREMENT 1,2021]=
ROUND(sales_amount[CURRENTV(),2020]*1.3,2)

RETURN UPDATED ROWS

```
SELECT product_type,year, month,sales_amount  
FROM sales
```

```
MODEL RETURN UPDATED ROWS
```

```
PARTITION BY product_type
```

```
DIMENSION BY (month,year)
```

```
MEASURES(sales_amount) (
```

```
Sales_amount[12,2020]=150)
```